

HIERARCHICAL GENERALIZED LINEAR MODELS WITH R. APPLICATIONS IN PLANT/ANIMAL BREEDING AND GENETICS

Lars Rönnegård

Högskolan Dalarna

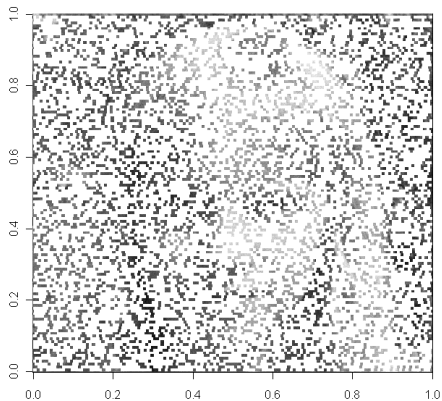
SLU

Sweden

Course outline

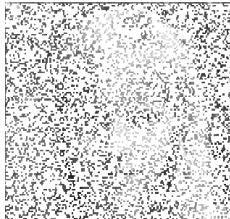
- **Linear regression**
 - diagnostic plots
- **Weighted least squares**
- **Generalized linear models**
- **Linear mixed models**
 - Estimating fixed and random effects
 - It's a weighted least squares problem!
 - Estimating variance components
 - Application on repeated observations
 - Fitting an animal model using the additive relationship matrix
 - Smoothing

Smoothing



The magic of smoothing

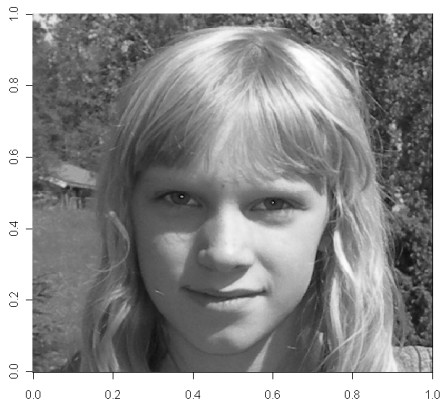
(b) 4×4 clusters are missing



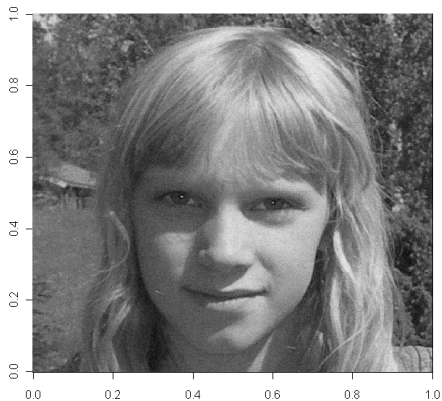
(d) Reconstruction of (b)



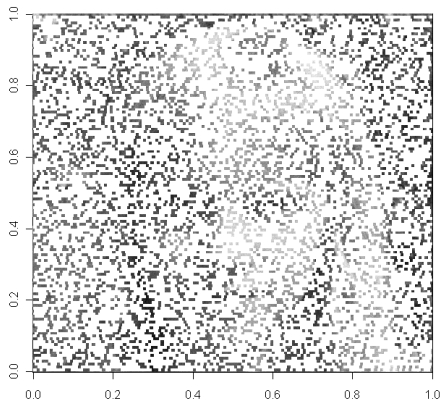
Original photo



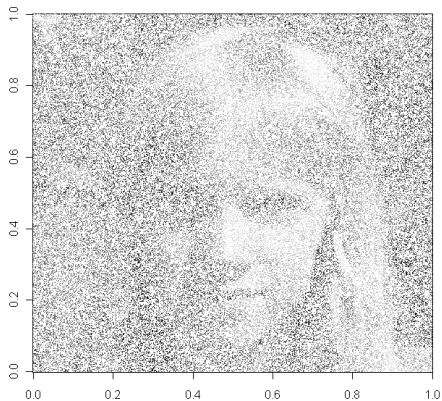
Added noise



4x4 deleted pixels

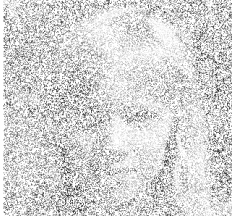


Deleted pixels at random

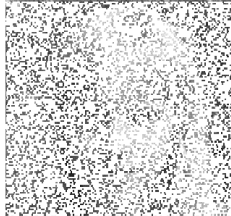


The magic of smoothing

(a) 70 % pixels are missing at random



(b) 4×4 clusters are missing



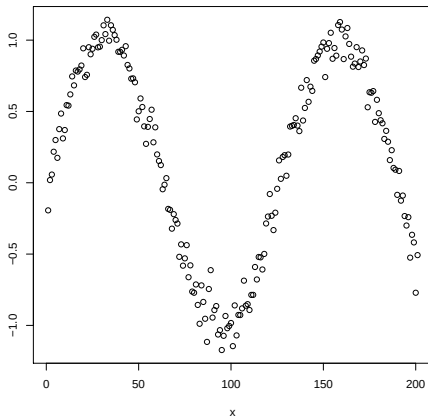
(c) Reconstruction of (a)



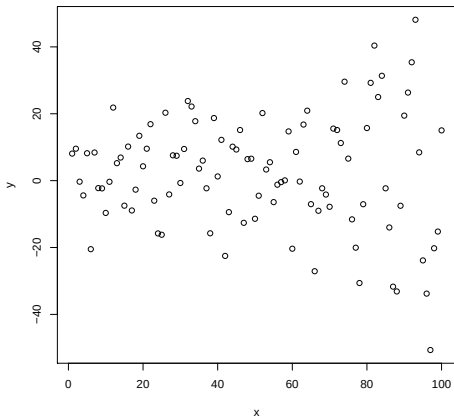
(d) Reconstruction of (b)



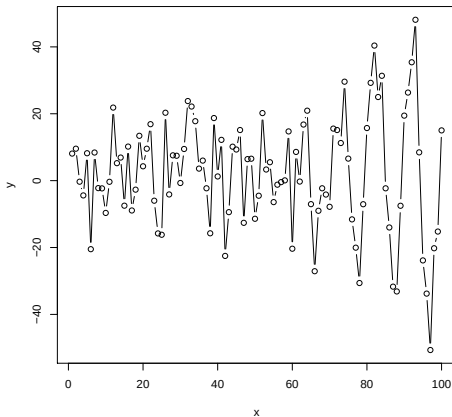
Smoothing of a function



How well can we smooth a very noisy function?



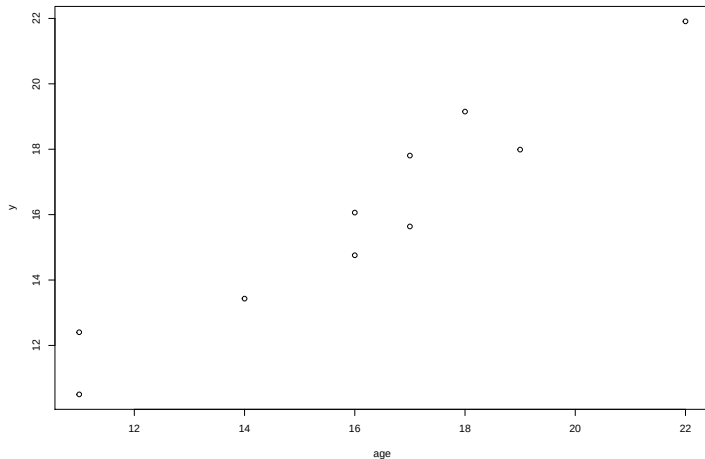
How well can we smooth a very noisy function?



Will be solved at the end of the course!

But let's start with linear regression.

Notation



Notation

y	age
18.0	19
19.2	18
17.8	17
21.9	22
15.6	17
14.8	16
16.1	16
10.5	11
12.4	11
13.4	14

$$y = X\beta + e$$

$$X = \begin{pmatrix} 1 & 19 \\ 1 & 18 \\ 1 & 17 \\ 1 & 22 \\ 1 & 17 \\ 1 & 16 \\ 1 & 16 \\ 1 & 11 \\ 1 & 11 \\ 1 & 14 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_{age} \end{pmatrix}$$

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Estimates

Call:

```
lm(formula = y ~ age)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.2274	-0.7024	-0.1721	0.8069	1.4200

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.6328	1.6688	0.379	0.714
age	0.9526	0.1016	9.375	1.37e-05 ***

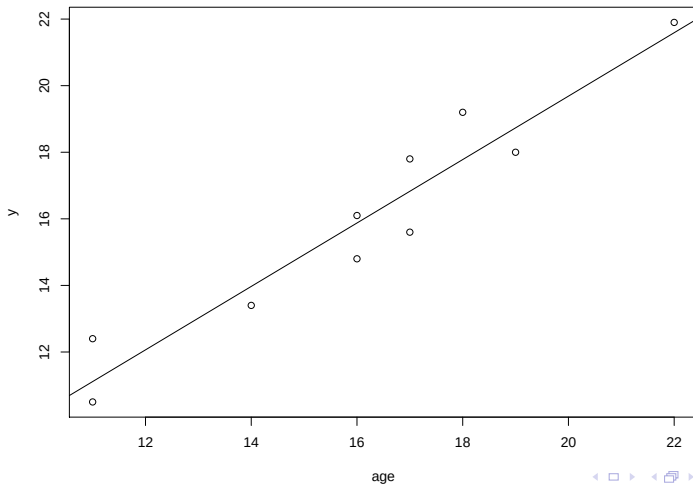
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.041 on 8 degrees of freedom

Multiple R-squared: 0.9166, Adjusted R-squared: 0.9061

F-statistic: 87.88 on 1 and 8 DF, p-value: 1.372e-05

Fitted regression line



Assumptions

$$\mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \\ e_{10} \end{pmatrix}$$

- The residuals e_i are normally distributed
- The residuals e_i have the same variance, i.e. $e_i \sim N(0, \sigma_e^2)$ for all i
- The residuals e_i are independently sampled, i.e. $\text{cov}(e_i, e_j) = 0$
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Estimating the regression coefficients

The regression coefficients β are fixed effects, because we assume that they have a fixed (but unknown) value.

They are estimated using least squares or maximum likelihood. Both methods give the same solution:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The least squares method minimizes the sum of squared residuals: $\sum e_i^2$.

In our simple example we have $f(\beta_0, \beta_{age}) = \sum (y_i - \beta_0 - \beta_{age} x_i)^2$

and the estimates are computed by finding the solution to:

$$\frac{df(\beta_0, \beta_{age})}{d\beta_0} = 0 \text{ and } \frac{df(\beta_0, \beta_{age})}{d\beta_{age}} = 0.$$

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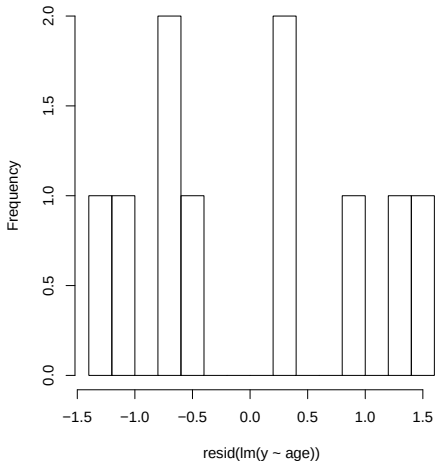
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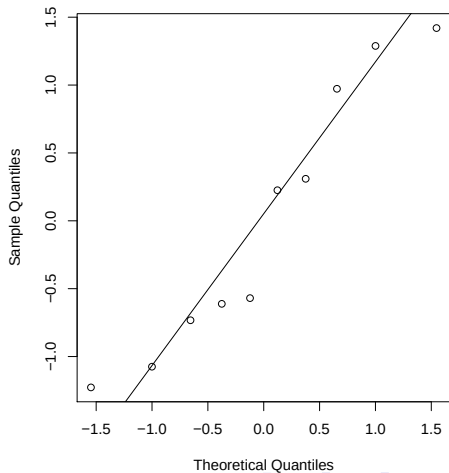
$$\frac{df(\beta_0, \beta_{age})}{d\beta_0} = 0 \text{ and } \frac{df(\beta_0, \beta_{age})}{d\beta_{age}} = 0.$$

Plot residuals

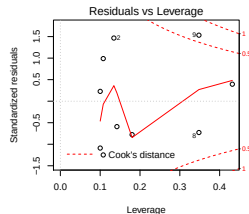
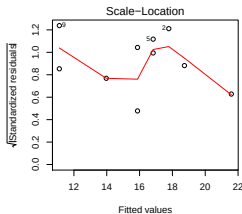
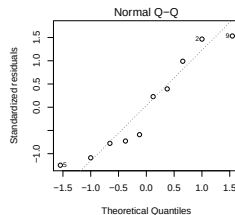
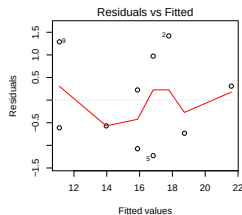
Histogram of residuals



Normal Q-Q Plot



Diagnostic plots



Terms used in diagnostic plots

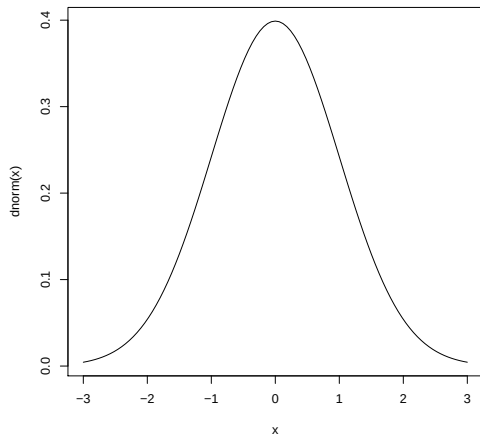
- **QQ-plot**
- **Leverages** (also known as hat values)
- **Standardized residuals**
- **Cooks distance**

QQ plot Theory

Construction

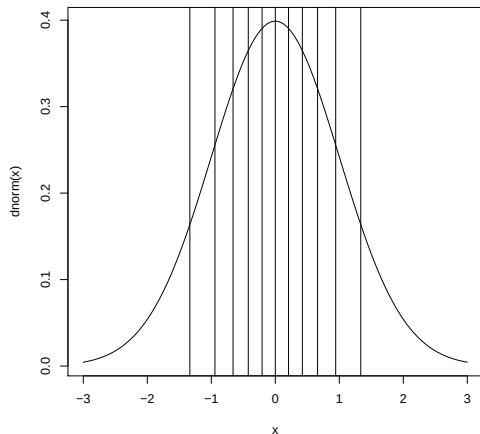
- Chop the normal density function into $n+1$ equally sized chunks
- The borders between each of the chunks are the values of the theoretical quantiles
- Plot the sorted observed values against the theoretical quantiles

QQ plot Theory

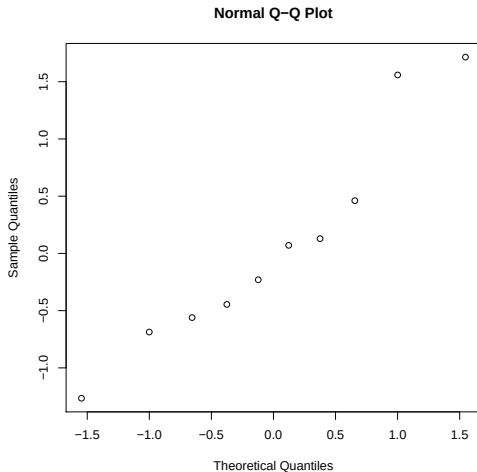


QQ plot Theory

Theoretical quantiles



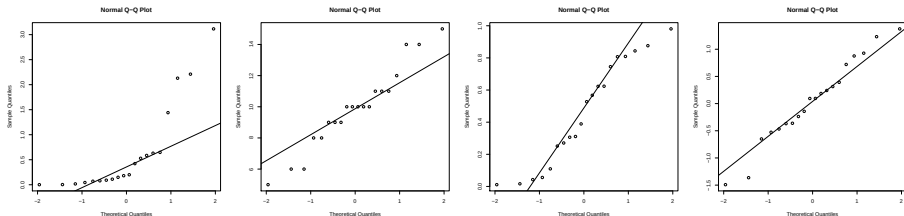
QQ plot Theory



Plots sorted observed values against the theoretical quantiles

Examples

Figure: 20 observations from a Chi-square ($df=1$) distribution, a Poisson distribution with mean 10, a uniform distribution and a normal distribution



The hat matrix

A matrix $\mathbf{H}$ that “puts the hat” on $\mathbf{y}$

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$$

where $\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$.

We have previously shown that $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, so

$$\mathbf{H} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

The use of the hat matrix is described in Hoaglin and Welsh (1978).

- A diagonal element h_{ii} measures how large proportion of the predicted value $\hat{y}_i$ depends on the i :th observed value y_i .
- The diagonal elements h_{ii} are called “leverages” or “hat values”
- The matrix $\mathbf{H}$ has a nice property (“idempotent”): $\mathbf{H} = \mathbf{H}\mathbf{H}$
- $h_{ii} = 1$; observation i adds no information to the likelihood
- $var(\hat{e}_i) = \frac{\sigma_e^2}{1-h_{ii}}$

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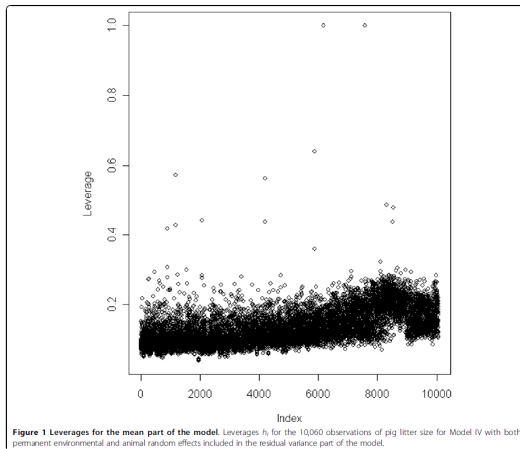
Hat values

Properties of hat values

- $var(\hat{e}_i) = \frac{\sigma_e^2}{1-h_{ii}}$
- h_{ii} bounded between $\frac{1}{n}$ and 1
- Rule of thumb: If there is a hat value twice as large as the second largest hat value, check it.

Hat values

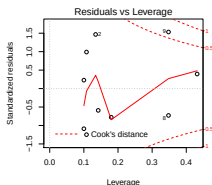
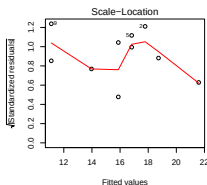
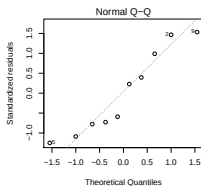
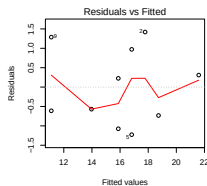
Rönnegård et al. (2010) GSE 42:8



Standardized residuals

Let $\hat{r}_i$ be standardized residuals so that $\text{var}(\hat{r}_i) = 1$ for all i . They are defined as:

$$\hat{r}_i = \frac{\hat{e}_i}{\hat{\sigma}_e \sqrt{1 - h_{ii}}}$$



Cook's distance

A measure of how influential an observation is on the results.

Cook's distance D_i of observation i is defined as the sum of all the changes in the regression model when observation i is removed from it

$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{p \hat{\sigma}_e^2}$$

Here p is the number of regression coefficients fitted in the model and $\hat{y}_{j(i)}$ is the fitted value for observation j when observation i has been removed from the data.

Fitting n different models to compute D_i could be time consuming. Is there a smarter way? Yes!

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Cook's distance

The hat values h_{ii} can be used to compute D_i .

$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{p\hat{\sigma}_e^2} = \frac{\hat{e}_i^2}{p\hat{\sigma}_e^2} \times \frac{h_{ii}}{(1 - h_{ii})^2}$$

So we do not need to fit n different models to compute D_i .

Leave-one-out cross-validation

Each time, Leave-one-out cross-validation (LOOCV) leaves out one observation, produces a fit on all the other data, and then makes a prediction at the x value for that observation that you left out.

The LOOCV statistic (aka Prediction Residual Sum of Squares, PRESS) is given by

$$CV = \frac{1}{N} \sum_{i=1}^N e_{[i]}^2$$

where $e_{[i]} = y_i - \hat{y}_{[i]}$ and $\hat{y}_{[i]}$ is the predicted value when the i :th observation is deleted.

But for a linear model we can simply compute it using the hat values

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Maximum likelihood estimate of the residual variance

Estimating the residual variance using maximum likelihood:

$$L(\beta_0, \beta_{age}, \sigma_e^2) = \prod \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_{age}x_i)^2}{2\sigma_e^2}\right)$$

$$\ell(\beta_0, \beta_{age}, \sigma_e^2) = \log L(\beta_0, \beta_{age}, \sigma_e^2)$$

$$= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma_e^2 - \frac{1}{2\sigma_e^2} \sum (y_i - \beta_0 - \beta_{age}x_i)^2$$

The solution to $\frac{d\ell(\beta_0, \beta_{age}, \sigma_e^2)}{d\sigma_e^2} = 0$, gives

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Estimating the residual variance using maximum likelihood:

$$L(\beta_0, \beta_{age}, \sigma_e^2) = \prod \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_{age}x_i)^2}{2\sigma_e^2}\right)$$

$$\begin{aligned} \ell(\beta_0, \beta_{age}, \sigma_e^2) &= \log L(\beta_0, \beta_{age}, \sigma_e^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma_e^2 - \frac{1}{2\sigma_e^2} \sum (y_i - \beta_0 - \beta_{age}x_i)^2 \end{aligned}$$

The solution to $\frac{d\ell(\beta_0, \beta_{age}, \sigma_e^2)}{d\sigma_e^2} = 0$, gives

$$\hat{\sigma}_e^2 = \frac{1}{n} \sum e_i^2$$

Unbiased estimation of the residual variance

Problem: The maximum likelihood estimator is biased

$$\hat{\sigma}_e^2 = \frac{1}{n} \sum e_i^2$$

$$E(\hat{\sigma}_e^2) \neq \sigma_e^2$$

In our first example we included the intercept and age in the model. It is possible to show that the maximum likelihood estimator in our example data set gives

$E(\hat{\sigma}_e^2) = \frac{n-2}{n} \sigma_e^2$. Hence, the following is an unbiased estimator

$$\hat{\sigma}_e^2 = \frac{1}{n-2} \sum e_i^2$$

For a model with p parameters in the model the unbiased estimator is

$$\hat{\sigma}_e^2 = \frac{1}{n-p} \sum e_i^2$$

The sum of hat values is equal to p . This is a relationship we will use later when we come to linear mixed models and define *effective number of parameters*.

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